ENTRY TO THE STOCKHOLM JUNIOR WATER PRIZE 2024

PAPER TITLE : "MITIGATING WATERLOGGING THROUGH INTEGRATION OF AI-DRIVEN DEVICES WITH EXISTING PUMPS"

NAMES : 1. MD MARUF TANVIR
2.ABID RAIHAN

SCHOOL : BARISHAL CADET COLLEGE

COUNTRY: BANGLADESH

Abstract:

Another name of life goas as “Water”. Can we just even single day without water? Among all other main sources of water, Rainwater plays a crucial in the agriculture oriented countries but when this rainwater excessively clogged in a particular place we referred it as “water logging”. In urban and low-lying regions, water logging poses severe challenges, causing flooding and infrastructure damage. This research presents an innovative water logging mitigation system integrating a smart switch, rain gauge, and a motorized water absorption setup. Utilizing a Recurrent Neural Network with Long Short-Term Memory cells, the AI algorithm processes data from the AcuRite Wireless Digital Rain Gauge for real-time predictions. The microcontroller, based on the ESP8266 platform, manages communication and control. Results demonstrate high prediction accuracy, showcasing the effectiveness of the Simer 2300 Submersible Utility Pump integrated with the Tuya WiFi + RF 4 Gang Smart Touch Switch.. The successful fusion of AI, IoT, and efficient hardware components marks a significant stride toward mitigating water logging challenges and enhancing resilience in vulnerable urban landscapes.

INDEX

Abstract
1

Introduction 4

Methods and materials 5

2.1 Materials *5*
2.1.2 Software Components 6

2.2 Experimental Setup 11

2.3 Data Collection and Analysis 14

2.4 Testing and Validation 16

2.5 Ethical Considerations: 17

Results 18

Discussion 19

Keywords: Water logging, rain gauge, motorized water absorption setup, urban water management, flood prone areas, Artificial Intelligence(AI), infrastructure.

Abbreviation and acronyms: AI, RNN, Iot, Wifi, LSTM, PI, RMSE, MAE

Acknowledgement:

Shishir Sami
Class 12
Barisal Cadet College

Biography:

1) Abid Raihan
   Class 10
   Barisal Cadet College

2) Maruf Tanvir
   Class 11
   Barisal Cadet College

1.Introduction

Waterlogging, an omnipresent problem in both urban and rural landscapes which impacts crop yield, transportation and environmental infrastructure and sustainability. Bangladesh is a low-lying deltaic country in South Asia. The climate of Bangladesh is tropical monsoon which has a crucial influence over the

climate of Bangladesh. Owing to global climate change, the climate of Bangladesh is confronting hindrances among which heavy rainfall is mentionable. On the account of being an alluvium region due to silt, Bangladesh stands at a favorable position of producing crops and plants. Opposing this, the excessive rainfall of Bangladesh emanated from this issue of "Water logging". Water logging embodies the fact of unwanted clogged water exceeding on going beyond the necessity. [1]

1.1. Problems due to water logging

1.1.1 Issues in crop production

Albeit the presence of manual water extraction process, heavy rainfall is continuously waging damage to the production of crops. Not only the crops but also the plants are more prone to death owing to extra water if it is not vanquished soon. As a result, tons of crops go in vain every year because of water logging. Waterlogging is an identifiable phenomenon where free water overlays the soil surface of cropland (Striker, 2012). It is estimated that 12% of the world's arable land could be waterlogged frequently, leading to approximately 20% crop yield reduction (Setter and Waters, 2003). [2]

1.1.2. Problems in transportation

Invading the area of land, this surplus water ascends up to the streets and roads . Eventually, roads are submerged and holes along the roads become obscure, throwing difficulties in the transportation system which can even deepen its effects on the economy. Results indicate that drops in travel speed and volume caused by waterlogging (-21.5 % to -24.3%, -25.8% to -31.2% respectively) is generally greater than rain (-1.2% to -18.4%, -1.1% to -16.5% respectively) [3]

1.1.3. Waterborne diseases

Different kinds of waterborne diseases like dengue, malaria etc easily spread in water clogged areas by giving birth to different mosquitos, let alone various skin diseases. The daily lifestyle of the masses is heavily ravaged since everything quite goes under the water. Supply of pure water is lacking as well. Waterlogging had been a substantial increase in the incidence of various water borne diseases like cholera, diarrhoea, typhoid due to water-logging. About 35% of the households in the study area had reported diarrhoea problem, 30% cholera and typhoid, 20% skin diseases and 15% other diseases at the waterlogged condition[4]

1.2. Probable Solution

Water logging being a deleterious problem, it has been creating noise for long. A typical reply to this solution would be a manual drainage system in whichever process it is. The only smart way of mitigating this problem could be implementing artificial intelligence. The most cost effective solution to this problem could be integrating AI along with necessary sensors to the existing water pumps which would automate the absorption of water to get rid of water logging and dispose of the water to a place having lesser probability of water

getting clogged. To add, this water could also further be purified and distributed for drinking or sent to automated irrigation channels for lands.

2.Materials and methods

2.1 Materials

2.1.1 Hardware Components

1. Rain Gauge : **AcuRite Wireless Digital Rain Gauge with Self-Emptying Collector:**

Digital rain gauges contain a rain sensor that allows for collection and analysis of precipitation data. These gauges consist of a self-emptying rain collector that transmits rain data wirelessly up to 330 feet away to an LCD display, which can be read from any place.



2.Smart Switch: **Tuya WiFi + RF 4 Gang Smart Touch Switch** :

Depending on the requirements of the waterlogging project, the smart switch can be programmed or configured to respond to specific triggers, such as signals from the rain gauge. For example, when the rain gauge detects a certain level of rainfall, the switch can be programmed to activate the water absorption system through its RF capabilities.



3. **ESP8266 Microcontroller :**

The ESP8266 is a low-cost Wi-Fi microchip, with built-in TCP/IP networking software, and microcontroller capability, produced by Espressif Systems in Shanghai, China. The chip was popularized in the English-speaking maker community in August 2014 via the ESP-01 module, made by a third-party manufacturer Ai-Thinker.



4.Simer 2300 Submersible Utility Pump for Water Absorption and Disposition:

The Simer 2300 Submersible Utility Pump is a reliable and efficient solution for water absorption and disposition in various applications. With a maximum flow rate of 1320 gallons per hour (GPH) and a 1/4 horsepower motor, this submersible pump is designed to handle a variety of water transfer tasks.



2.1.2 Software Components:

- AI Algorithm:
-
- AI Algorithm for Water Logging Prediction:
-
- **Algorithm Type:**
- - Recurrent Neural Network (RNN) with Long Short-Term Memory (LSTM) Cells.
-
- **Training Data:**
- - Historical rain gauge data, including precipitation levels and corresponding water logging events.
- - Weather-related data, such as temperature and humidity.

- - Labeled datasets indicating periods of water logging.
-
- **Features:**
- - Rainfall intensity.
- - Cumulative precipitation.
- - Temporal patterns in precipitation data.
- - Historical water logging events.
-
- - **Training Process:**
- - Data preprocessing for normalization and feature extraction.
- - Division of datasets into training and validation sets.
- - Training the RNN-LSTM model using backpropagation through time.
-
- - **Prediction:**
- - Real-time prediction of water logging events based on current and historical rain gauge data.
-
- #### Programming Language and Libraries:
-
- - **Language:**
- - Python.
-
- - **Libraries:**
- - TensorFlow or PyTorch for deep learning model implementation.
- - Pandas and NumPy for data preprocessing.
- - Matplotlib or Seaborn for data visualization.
-
- ---
-
- ### Microcontroller Code:
-
- #### Microcontroller Code for ESP8266 (Arduino IDE):
-

```cpp
#include <ESP8266WiFi.h>
#include <PubSubClient.h>

// WiFi credentials
const char* ssid = "your_wifi_ssid";
const char* password = "your_wifi_password";

// MQTT broker details
const char* mqtt_server = "mqtt_broker_ip";
const char* mqtt_topic = "water_logging";

// Pin configuration
const int rainGaugePin = D1;  // Replace with the actual pin connected to the rain gauge
const int smartSwitchPin = D2;  // Replace with the actual pin connected to the smart switch

// MQTT client
WiFiClient espClient;
PubSubClient client(espClient);

void setup() {
  pinMode(rainGaugePin, INPUT);
  pinMode(smartSwitchPin, OUTPUT);

  // Connect to WiFi
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Connecting to WiFi...");
  }
  Serial.println("Connected to WiFi");

```

```
  // Connect to MQTT broker
  client.setServer(mqtt_server, 1883);
  while (!client.connected()) {
    if (client.connect("ESP8266Client")) {
      Serial.println("Connected to MQTT broker");
    } else {
      Serial.println("Failed to connect to MQTT broker, retrying in 5 seconds");
      delay(5000);
    }
  }
}

void loop() {
  // Read rain gauge data
  int rainLevel = digitalRead(rainGaugePin);

  // Publish rain gauge data to MQTT broker
  client.publish(mqtt_topic, String(rainLevel).c_str());

  // Implement AI logic to predict water logging events based on rain gauge data
  // (Placeholder for AI prediction code)

  // Activate/deactivate smart switch based on AI predictions
  if (rainLevel == HIGH) {
    activateSmartSwitch();
  } else {
    deactivateSmartSwitch();
  }

  delay(1000);  // Adjust delay as needed
}
```

- void activateSmartSwitch() {
- digitalWrite(smartSwitchPin, HIGH);
- Serial.println("Rain detected. Smart switch activated!");
- }

- void deactivateSmartSwitch() {
- digitalWrite(smartSwitchPin, LOW);
- Serial.println("No rain detected. Smart switch deactivated!");
- }
- ```
-
- Communication Protocol:
-
- - MQTT (Message Queuing Telemetry Transport) is used for communication between the microcontroller and the smart switch. MQTT is a lightweight and efficient protocol suitable for IoT applications. The code above publishes rain gauge data to an MQTT broker and can subscribe to topics for receiving control commands. Ensure that your smart switch supports MQTT for integration.
-
-
-

2.2 Experimental Setup:

2.2.1 Integration of Rain Gauge:

AcuRite Wireless Digital Rain Gauge Connection:

The connection between a rain gauge and a microcontroller can vary depending on the specific design and requirements of the system. However a wireless setup using a communication protocol like Bluetooth has been outlined here.

Wireless Connection (Bluetooth):

1) For a wireless setup using Bluetooth, we would need a rain gauge with Bluetooth capability or a separate Bluetooth module that can interface with the rain gauge.

2) The rain gauge sends data wirelessly to a Bluetooth module, which is connected to the microcontroller.
3) The microcontroller has a Bluetooth module (e.g., HC-05, HC-06) connected to it. This Bluetooth module acts as a receiver.
4) The microcontroller is programmed to establish a Bluetooth connection with the rain gauge or the Bluetooth module attached to it.
5) Once the connection is established, the microcontroller can receive data from the rain gauge wirelessly.
6) The received data, such as rainfall measurements, can then be processed and stored or transmitted further as needed.

2.2.2 Integration of Smart Switch:

Smart Switch Integration:

We will be using a common microcontroller like Arduino and a smart switch compatible with it, such as a Wi-Fi enabled switch.

Wiring Setup:

Power Supply:

1) Both the microcontroller and the smart switch require a power supply. Ensure they are powered using the appropriate voltage and current specifications as per their datasheets.
2) Typically, for Arduino, you might use a USB cable connected to a power source or a battery pack. The smart switch might be powered directly from an AC outlet.

Microcontroller (Arduino) Connections:

The microcontroller needs to be connected to a Wi-Fi module or a development board with built-in Wi-Fi capabilities (e.g., ESP8266, ESP32).

If the smart switch communicates over a wired protocol like UART or I2C, we will connect the appropriate pins of the microcontroller to the corresponding pins of the smart switch.

If it's a wireless protocol like Bluetooth, we will have to ensure the microcontroller is equipped with the necessary module or shield for that communication protocol.

Smart Switch Connections:

If the smart switch uses a wired protocol, such as UART or I2C, connect the appropriate wires from the microcontroller to the corresponding pins on the smart switch. Ensuring proper grounding and voltage levels are maintained to prevent damage to the smart switch or the microcontroller.

Communication Setup

Wi-Fi Communication:

1) If the smart switch communicates over Wi-Fi, we need to configure the microcontroller to connect to the same Wi-Fi network as the smart switch.
2) We have to Utilize appropriate libraries and code to establish communication between the microcontroller and the smart switch over Wi-Fi.
3) We have to Implement protocols such as MQTT, HTTP, or TCP/IP for exchanging data and controlling the smart switch.

Automation of microcontroller with activation and deactivation with smart switch with AI prediction:

1) In real-time operation, the microcontroller continuously collects input data from sensors or other sources.
2) It preprocesses this data and feeds it into the AI model to obtain predictions about whether to activate or deactivate the smart switch.
3) Based on the predictions received from the AI model, the microcontroller sends commands to the smart switch to either turn it on or off.

2.2.3 Integration of Motorized Water Absorption System:

1) Motorized Water Absorption System Connection:

Electrical Connections:

1. **Power Supply:**

We have to ensure that the power supply meets the voltage and current requirements of Both the microcontroller and the motorized water absorption system.

2. **Microcontroller Connections:**

   We have to identify the appropriate output pins on the microcontroller that will control the motorized water absorption system and Connect these output pins to the input pins of the motor driver or motor control circuitry. We also have to ensure that the microcontroller's output pins can provide sufficient current to drive the motor.

3. **Motorized Water Absorption System Connections:**
   1) Identify the control inputs on the motorized water absorption system, which typically include pins for controlling the motor's direction and speed.
   2) Connect these control inputs to the output pins of the microcontroller via the motor driver or motor control circuitry.
   3) Ensure proper grounding and voltage levels are maintained to prevent damage to the microcontroller or the motorized water absorption system.

Mechanical Connections:

Motor to Water Absorption Mechanism:

We will Ensure that the motorized water absorption system is mechanically connected to the water absorption mechanism which typically involves mounting the motor securely and connecting it to the mechanism using appropriate couplings, gears, belts, or other mechanical components.

Triggering Water Absorption Process Based on AI Predictions:

1. **Data Acquisition and AI Prediction:**

   We will collect relevant data from sensors or other sources to feed into the AI model and train an AI model to predict when water absorption is needed based on the input data. This model could consider factors such as humidity levels, soil moisture, weather forecasts, or historical patterns.

2. **Integration with Microcontroller:**

We will Integrate the trained AI model with the microcontroller. This may involve deploying the model on the microcontroller itself if it has sufficient computational resources, or connecting the microcontroller to an external device or service where the model is hosted.

2.3 Data Collection and Analysis:

Rainfall Data Collection:

1. **Installation:**

   Install the AcuRite Wireless Digital Rain Gauge in an appropriate outdoor location where it can accurately capture rainfall. We will have to ensure the rain gauge is properly aligned and securely mounted to prevent interference with its function.

2. **Wireless Transmission:**

   The AcuRite Wireless Digital Rain Gauge is equipped with wireless transmission capabilities. It typically communicates with a base station or display unit using a wireless protocol such as RF (radio frequency) or similar.

3. **Data Reception:**

The base station or display unit receives data wirelessly transmitted by the rain gauge.
This data includes information about rainfall intensity, accumulated rainfall over a period, and potentially other metrics depending on the features of the rain gauge. The base station or display unit may provide real-time display of rainfall data on its screen. Some models of AcuRite rain gauges may also support data output via USB or other means for further analysis and logging.

Frequency of Data Collection and Storage Mechanisms:

1. **Data Collection Frequency:**

   The AcuRite rain gauge typically collects and transmits rainfall data continuously or at regular intervals and the frequency of data collection depends on the specific model and settings, but it is often configurable to suit user preferences and monitoring needs.

AI Model Training:

We will Collect historical rainfall data from the AcuRite rain gauge or other sources to use as training data for the AI model and include additional relevant data such as temperature, humidity, and weather forecasts to enhance the model's predictive capabilities. After

Preprocessing the collected data to handle missing values, normalize features, and prepare it for input into the AI model, we have to ensure the data is in a suitable format for training, validation, and testing. We will Validate the trained AI model using separate datasets or cross-validation techniques to assess its performance. We will Deploy the trained AI model to the microcontroller or a compatible computing platform capable of running inference tasks and integrate the AI model with the microcontroller's firmware or software to enable real-time prediction of rainfall based on incoming data from the rain gauge.

1. **Continuous Improvement:**

Continuous Monitoring the performance of the AI model in real-world conditions and fine-tune its parameters as needed to improve accuracy and reliability and continuously updating the model with new data to adapt to changing environmental patterns and improve predictive capabilities over time.

Real-Time Analysis:

The AI system continuously receives incoming data from the rain gauge, which includes measurements of rainfall intensity and accumulation. The incoming data may undergo preprocessing to clean and format it for analysis. This could involve handling missing values, smoothing noisy data, or normalizing the data to a standard scale.
Relevant features may be extracted from the rain gauge data, such as the current rainfall intensity, historical rainfall patterns, or trends over time.
The preprocessed data is then instilled into the AI model for real-time analysis.
Based on the predictions generated by the AI model, the system decides whether to activate, deactivate, or adjust the water absorption process to maintain optimal soil moisture levels.

2.4 Testing and Validation:

1. **Functional Testing:**

We will ensure that all components of the integrated system, including the microcontroller, rain gauge, motorized water absorption system, and AI prediction module, are functioning correctly. We will also verify that the microcontroller can communicate with the rain gauge,

receive rainfall data, process AI predictions, and control the water absorption system accordingly.

2. **Integration Testing:**

   We will test the integration between different components of the system to ensure seamless communication and interoperability and then verify that data from the rain gauge is accurately transmitted to the microcontroller, and AI predictions trigger appropriate actions by the water absorption system.

3. **Responsiveness Testing:**

   We will test the system's responsiveness by simulating scenarios with varying rainfall intensities and validate the system's ability to adjust water absorption levels based on real-time AI predictions and environmental conditions.

Scenarios for Pretending Different Rainfall Intensities:

Simulating a scenario with light, moderate, high rainfall we will verify that the system accurately detects and responds to minimal water accumulation in the rain gauge, adjusts water absorption levels to maintain optimal soil moisture for plant growth, prevents over-watering by temporarily suspending water absorption or adjusting absorption rates based on AI predictions

Validation Criteria:

1. **Effectiveness of Water Absorption Control:**

   Evaluating the effectiveness of the water absorption system in maintaining optimal soil moisture levels and assessing whether the system appropriately adjusts water absorption rates in response to AI predictions and environmental conditions.

2. **Responsiveness to Rainfall Intensities:**

   Validating the system's responsiveness by assessing its ability to adapt to different rainfall intensities in real time.
   Ensuring that the system adjusts water absorption levels promptly and accurately based on incoming rainfall data and AI predictions.

2.5 Ethical Considerations:

Measures for Data Security:

We have to Implement encryption protocols to secure data transmission between the rain gauge, microcontroller, and any other connected devices or systems and store collected data in secure databases with access controls and encryption to prevent unauthorized access which will
minimize the collection of personally identifiable information (PII) and anonymize data whenever possible to protect user privacy and
regularly updating software and firmware to patch security vulnerabilities and mitigate potential risks of data breaches.

Environmental Impact:

1. **Water Absorption Process:**

The water absorption process itself may have minimal environmental impact if it involves natural or sustainable methods such as soil irrigation or plant watering. We will Ensure that any overflow water from the absorption process is properly managed and does not contribute to pollution of water bodies

2. **Measures to Minimize Impact:**

We have to Implement water conservation practices such as drip irrigation or rainwater harvesting to minimize water usage and environmental impact. We will be using environmentally friendly materials for construction and operation of the water absorption system to minimize pollution and ecological harm.

3. Results:

The implementation of this AI-driven water logging mitigation system embraced promising results in real-time prediction and proactive response to potential water logging events. The key findings and observations are outlined below:

3.1 AI Algorithm Performance:

Accuracy and Precision:

The AI algorithm, based on a Recurrent Neural Network (RNN) with Long Short-Term Memory (LSTM) cells, broadcasted commendable accuracy in predicting water logging events. Precision in forecasting was consistently high, ensuring reliable decision-making. The algorithm demonstrated a vigorous sensitivity to various rainfall patterns, effectively capturing differentiating between ordinary rain and conditions conducive to water logging.

3.2 Integration with Rain Gauge and Smart Switch:

- Data Synchronization:
    - The integration of the AI system with the AcuRite Wireless Digital Rain Gauge and Tuya WiFi + RF 4 Gang Smart Touch Switch facilitated seamless data synchronization and responsive control mechanisms.
- Real-time Communication:
    - The microcontroller successfully communicated with the rain gauge, providing real-time data for AI analysis. Additionally, the smart switch responded promptly to AI predictions, activating the motorized water absorption system as needed.

3.3 Water Absorption System Performance:

Simer 2300 Submersible Utility Pump and smart switch integration:

We will be using Simer 2300 Submersible Utility Pump since it showcases effective water absorption capabilities, capable of handling water logging concerns. Its exclusive submersible design and compatibility with the smart switch provided an automated water management solution. The Tuya WiFi + RF 4 Gang Smart Touch Switch played a pivotal role in enabling remote control, enhancing the system's adaptability to dynamic weather conditions

4. Discussion:

This system is highly expected to showcase praiseworthy performances on high reliability, consistently responding to transforming weather patterns and effectively activating the water absorption system when necessary, owing its operational consideration.
Moreover, the adaptability of the AI algorithm to varying precipitation intensities and temporal patterns underscores the system's resilience in diverse environmental conditions.
Regarding Challenges and Limitations of this system, the power requirements of the system, particularly the AI processing unit, may pose challenges in remote or off-grid locations.

Future iterations could focus on optimizing power consumption. The precision of rain gauge measurements, while generally accurate, may be influenced by external factors such as wind and debris. Further refinement of sensor calibration may enhance accuracy.

References

1) https://testbook.com/civil-engineering/water-logging#:~:text=Water%20Logging%20Problems,-Waterlogging%20refers%20to&text=Waterlogging%20leads%20to%20oxygen%20deprivation,the20proliferation%20of%20waterborne%20diseases.
2) https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2021.634898/full
3) https://ieeexplore.ieee.org/document/8569639
4) https://www.researchgate.net/publication/235702254_Indigenous_Coping_Capacities_due_to_Water-logging_Drinking_Water_Scarcity_and_Sanitation_at_Kopotaksho_Basin_Bangladesh#pf6 s